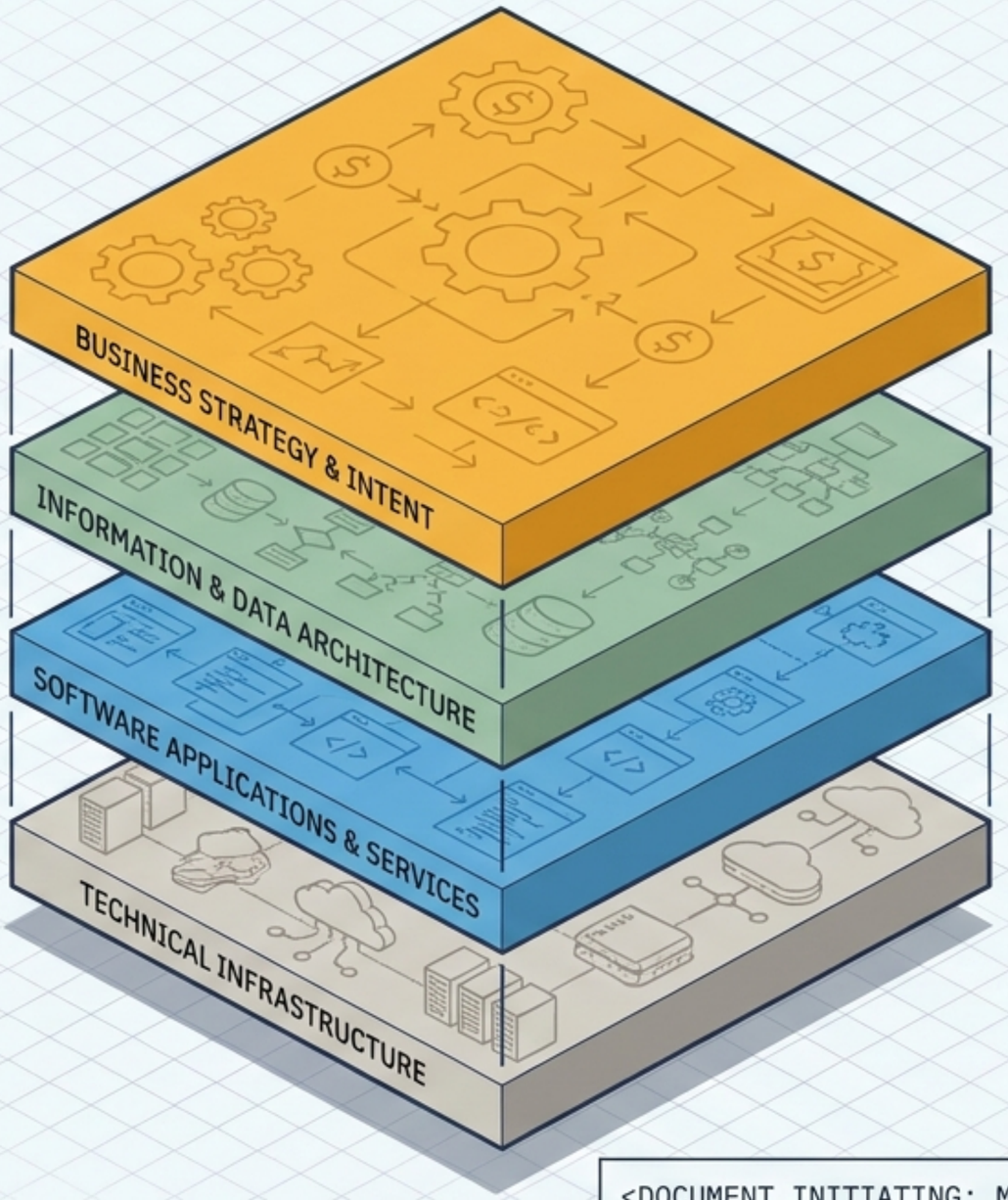
The Solution Architect’s Blueprint

Bridging Business Intent with Technical Reality



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The Architecture Ecosystem

	SCOPE & HORIZON	PRIMARY FOCUS	INFORMATION FLOW
Enterprise Architect	Multi-year, Organization-wide	Strategy, target-state roadmap, tech standards	Top-down guardrails
Business Architect	Strategic alignment	Capabilities, processes, business benefits	Sideways context for "why"
Technical (Domain) Architect	Deep specialization	Security, network, cloud, data mechanics	Sideways specialist validation
Solution Architect	Single project lifecycle	End-to-end coherence, delivering business needs	Translator: Bridges top-down rules and sideways expertise

Solution Architects act as the translator—taking top-down guardrails and sideways domain expertise to build coherent, bottom-up solutions, feeding exceptions back to refine the enterprise roadmap.

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Business Architecture: The “Why” and the “What”

[DOMAIN: BUSINESS]

The Goal
(Strategy Map & Canvas)

Financial (Revenue/Cost)	Customer (Retention)
Internal Processes (Efficiency)	Learning & Growth

- How the business creates value.

The Engine
(Capability Model)

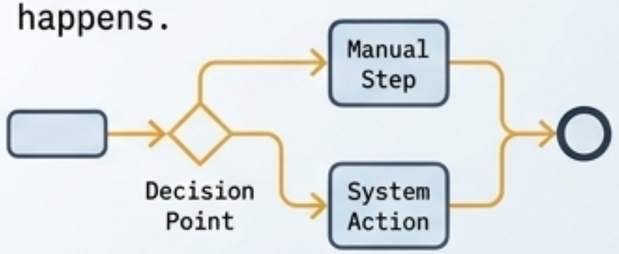
- What the business does independent of tech/org.
- E.g., “Process claims”



- Used to prioritize investment and spot gaps or duplication.

The Flow
(Process Description)

- How the work happens.



- Identifies manual pain points, handoffs, and direct system requirements.

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The Benefits Dependency Network

[DOMAIN: BUSINESS]



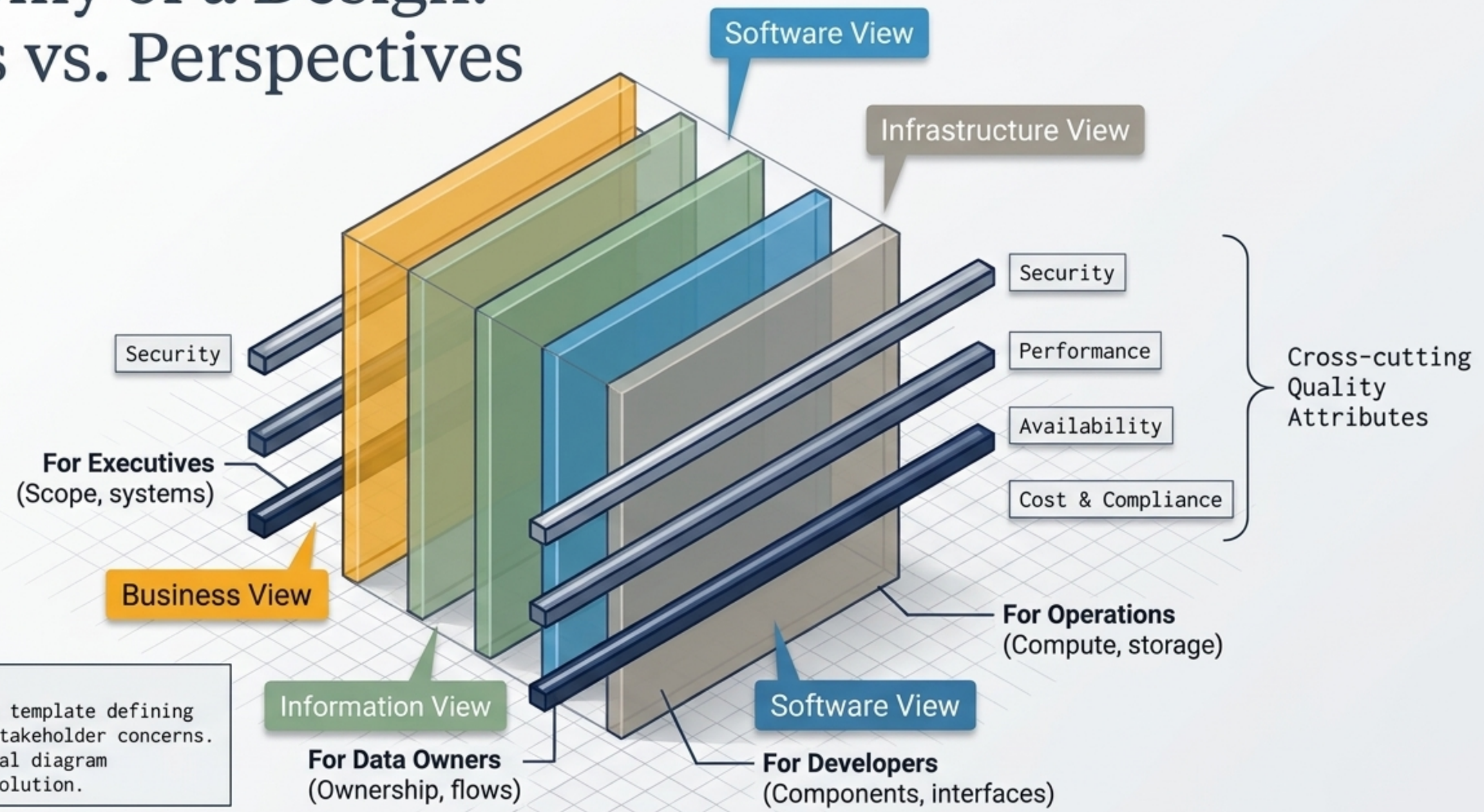
Traceability is the filter: If a tech component traces to no benefit, question it. If a benefit has no supporting change, it will never be realized.

Cost Justification Checklist

1. ROI (Return on Investment)
2. Payback Period
3. NPV (Net Present Value)
4. TCO (Total Cost of Ownership)

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Anatomy of a Design: Views vs. Perspectives



Terminology Note:
A **Viewpoint** is the template defining the notation and stakeholder concerns.
A **View** is the actual diagram produced for the solution.

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Architecture Analysis & The ARB Process

Phase 1: Architecture Analysis (The Stress Test)

Trade-off Mapping: [BUSINESS]
Balancing opposing qualities (e.g., Security vs. Convenience)

Scenario Evaluation: [INFORMATION]
"What happens when traffic triples?"

Risk & Gap Analysis: [SOFTWARE]
Identifying untested assumptions

Prototypes: [INFRASTRUCTURE]
Proving the riskiest parts early

Phase 2: The ARB Cycle



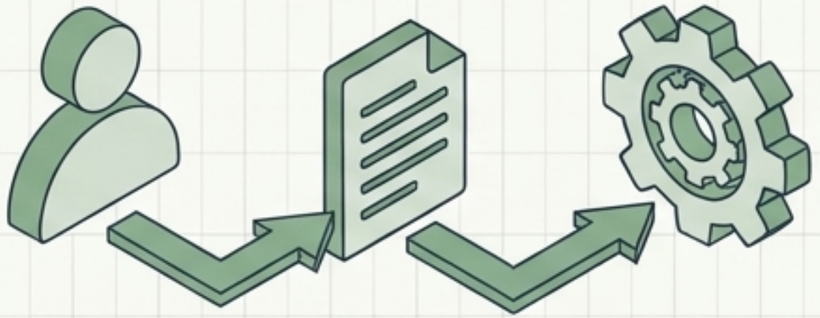
The Architecture Review Board (ARB) is not a roadblock. It is a governance safety net designed to catch enterprise-level risks that a single project team might miss.

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Information Architecture vs. Data Architecture

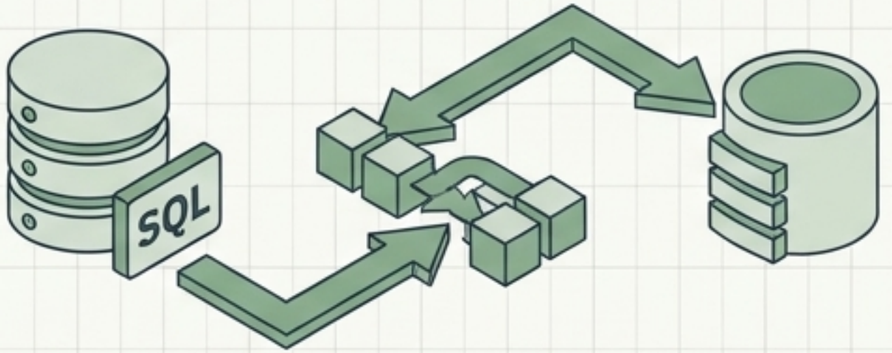
[DOMAIN: INFORMATION]

Information Architecture (Business-Facing)



Focuses on business meaning, flow, and value (e.g., Defining universally what a ‘Customer’ or ‘Claim’ is across the organization).

Data Architecture (Technical-Facing)



Focuses on technical implementation: conceptual, logical, and physical models, storage, warehouses, and data lakes.

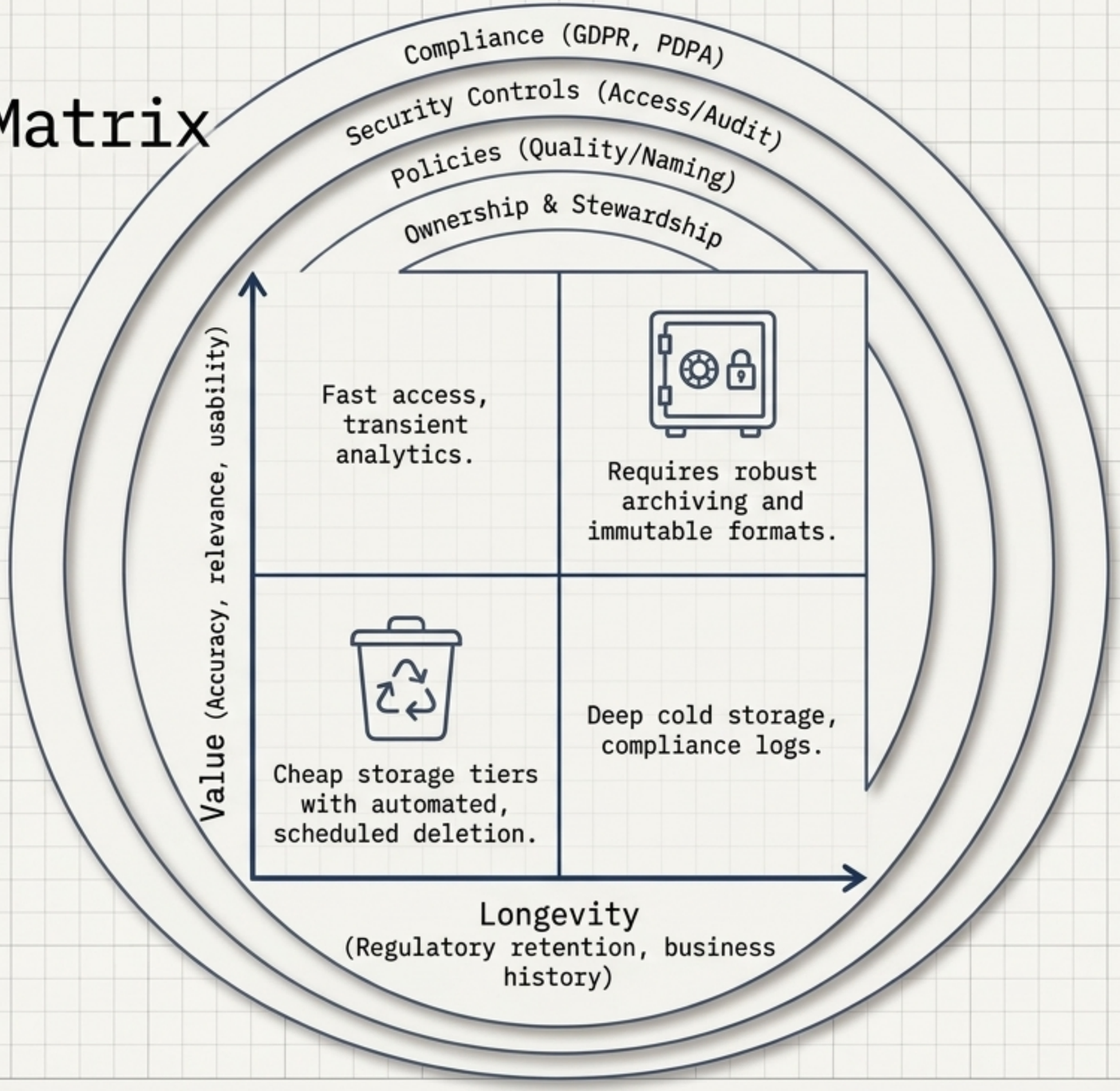
Open Source Enablers (Driving portability & lowering lock-in)

- **Frameworks:** TOGAF, DAMA-DMBOK
- **Schemas:** schema.org standards
- **Storage/Streaming:** PostgreSQL, Kafka
- **Portable Formats:** JSON, Parquet, CSV

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The Data Lifecycle Matrix

[DOMAIN: INFORMATION]



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Software Architecture: Patterns & The Cloud

[DOMAIN: SOFTWARE]

Principles of Good Design

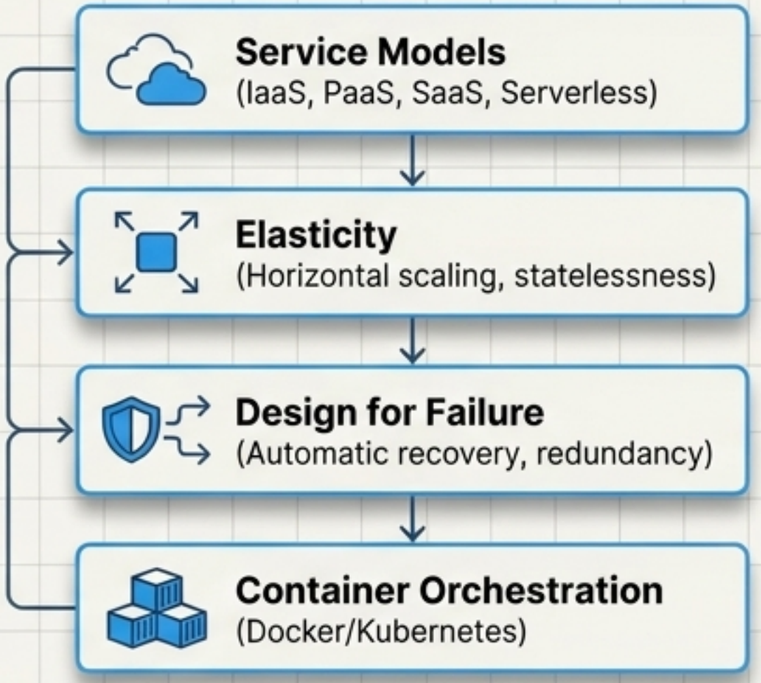
Separation of Concerns:
One clear job.

Loose Coupling:
Interface-dependent.

High Cohesion:
Related functions stay together.

Reuse:
Build once, share universally.

The Cloud Mindset Blueprint



The MVP Tradeoff

Speed to Market — Technical Debt

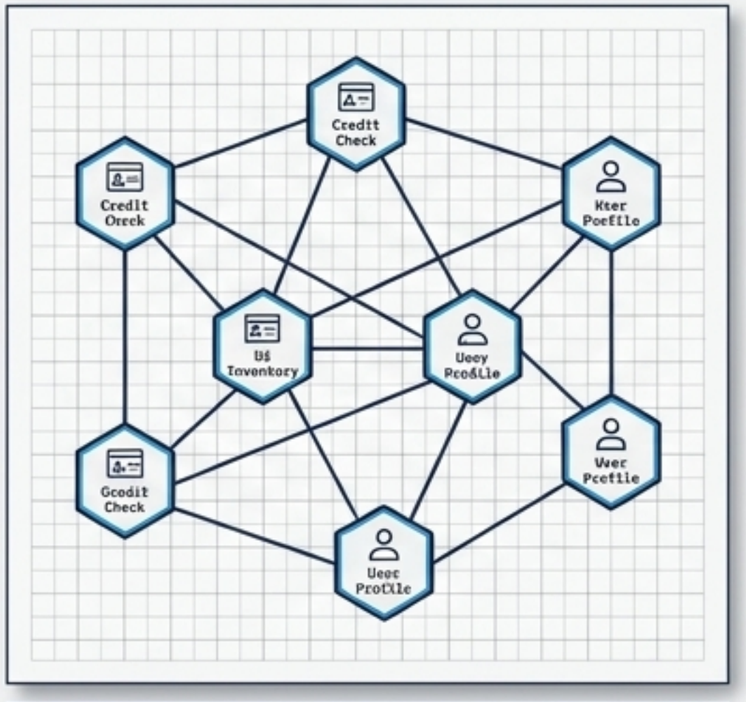
Define the core to reduce risk early. Use managed services to move fast, but design clean interfaces for future growth without rebuilding.

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Integration: The Nervous System

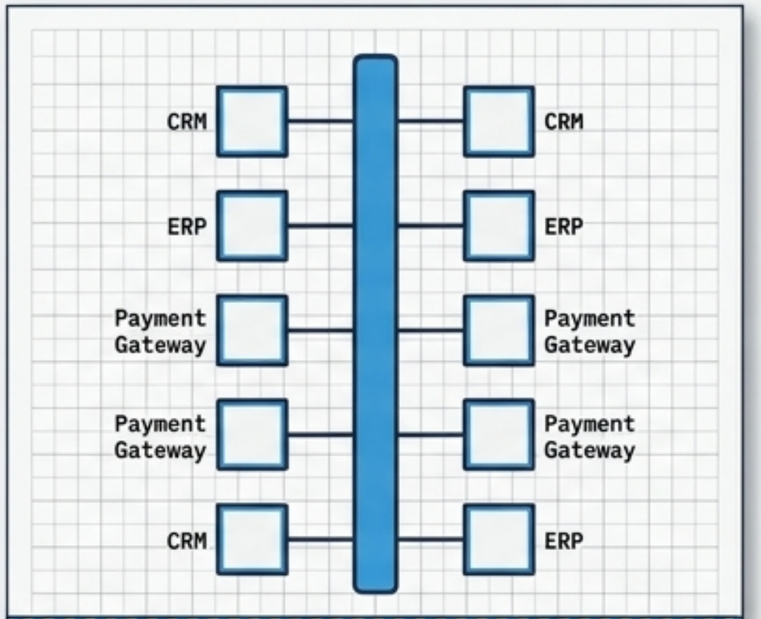
[DOMAIN: SOFTWARE]

Pattern 1: SOA (Service-Oriented)



Reusable business functions (e.g., 'check credit').
Pro: Flexibility.
Con: Tangle of dependencies without governance.

Pattern 2: ESB (Enterprise Service Bus)



Warning: Risk of becoming a central bottleneck/single point of failure.

Central backbone for routing, protocol mediation, and transformation (XML to JSON).

Pattern 3: Mashups



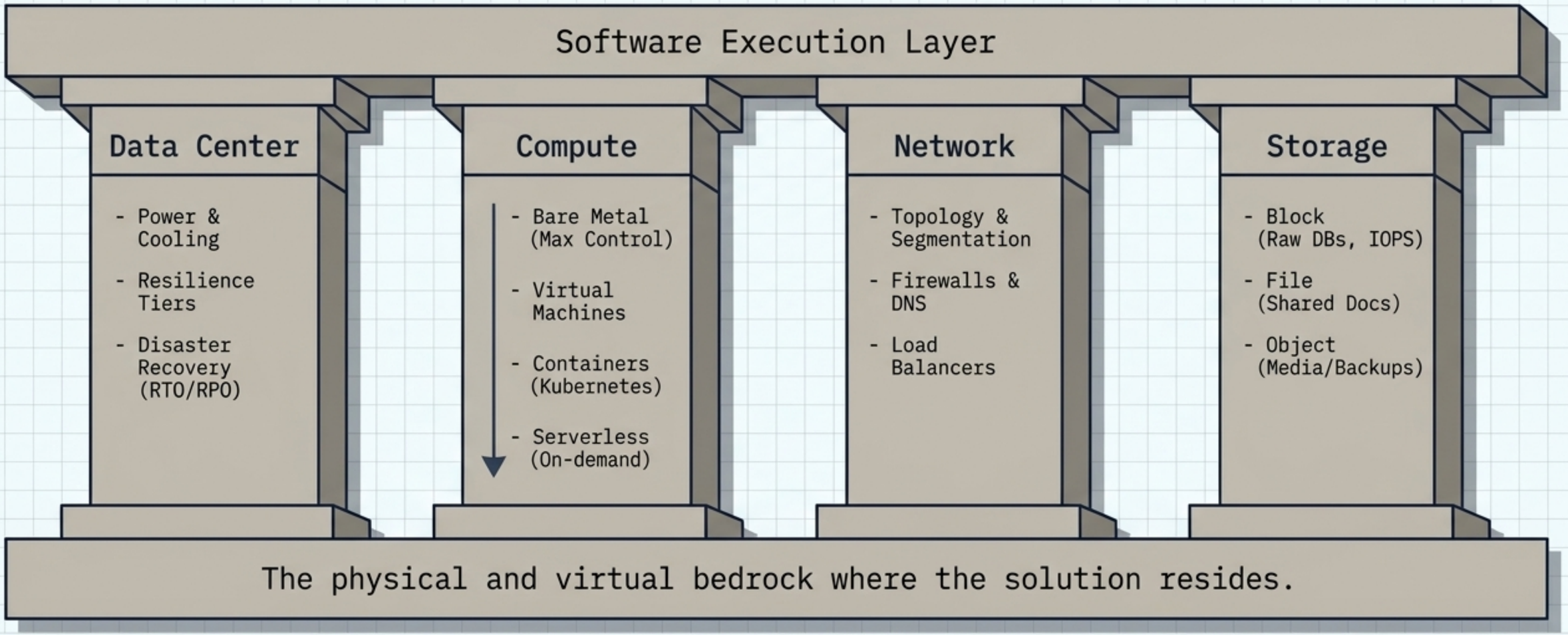
Warning: Beware of external dependencies and shadow IT.

Lightweight, rapid assembly of external APIs and UI data.

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Infrastructure Architecture: The Foundation

[DOMAIN: INFRASTRUCTURE]



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The Solution Lifecycle Engagement

Idea/Design

`[ROLE: HIGH ENGAGEMENT]`

- Shape scope and analyze options.
- Secure ARB approval.
- Clarify architecturally significant requirements.

Build/Deploy

`[ROLE: GUARDIAN]`

- Guide development team.
- Protect design from shortcuts.
- Track technical debt and integration.

Operate/Retire

`[ROLE: ADVISORY]`

- Verify business benefits.
- Monitor live performance.
- Plan decommissioning.

The architect's job doesn't end at the design document. They own the integrity of the solution from **idea** to retirement.

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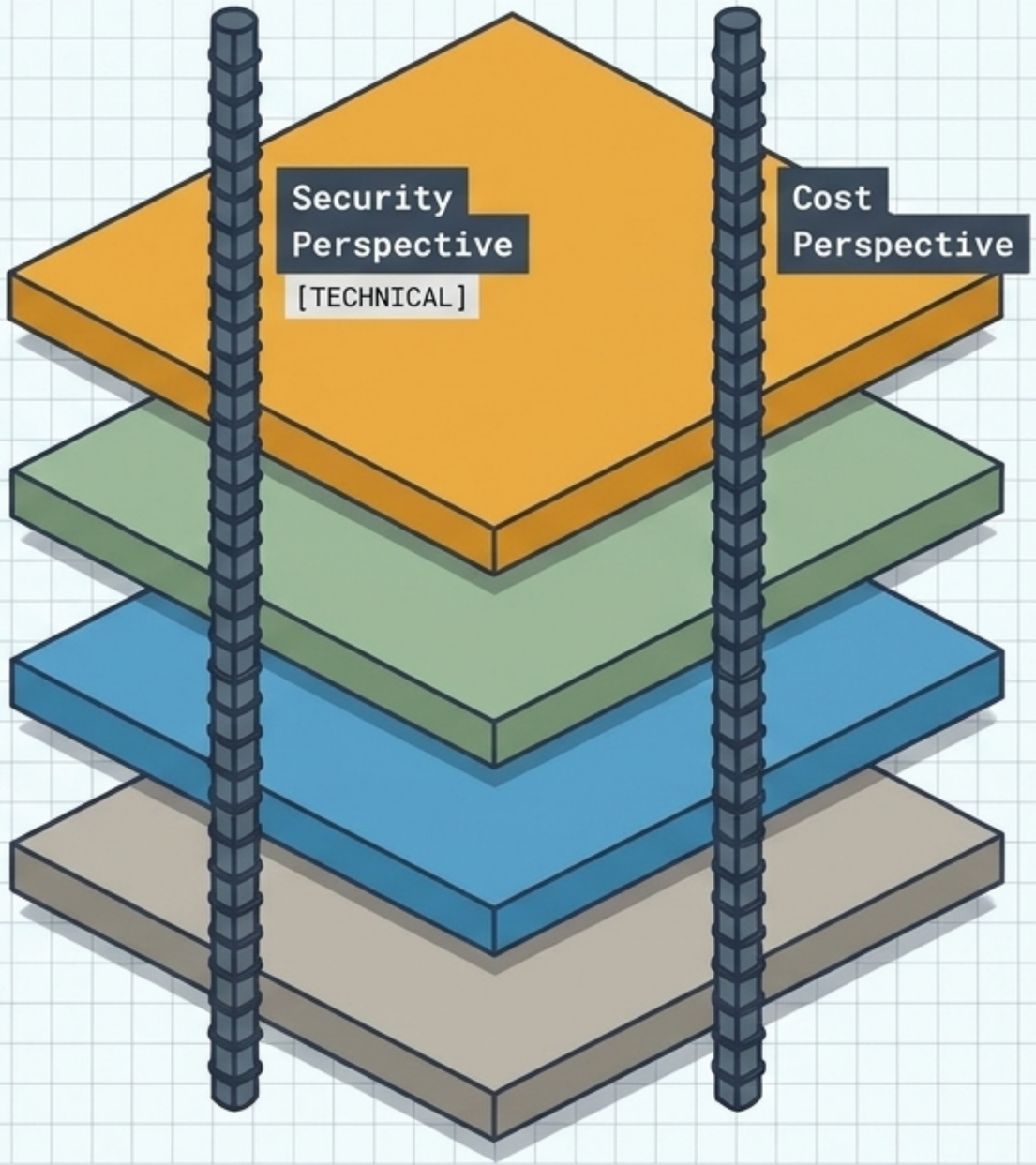
Synthesis: The Complete Architecture Stack

Business View
Capability models & process flows.
(Answers: What are we achieving?)

Information View
Master data models & lifecycles.
(Answers: What is it worth & how is it structured?)

Software View
Components, APIs & ESB integration.
(Answers: How is it built & connected?)

Infrastructure View
Compute clusters, segmented networks, block storage.
(Answers: Where does it run?)



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Summary Checklist: The 8 Coverage Areas

1. Business Architecture (Capabilities & Benefits)
2. Information & Data Architecture
3. Software & Application Architecture
4. Infrastructure & Cloud Architecture
5. Integration & System Topologies
6. Security & Compliance Controls
7. Operations & Service Management
8. Cost, Risk & Governance

A successful Solution Architect owns the integrity of the solution from idea to retirement. They must combine the creativity of a designer, the rigor of an engineer, and the strategy of a business executive.

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